Questionnaire for Expert Opinion Survey on Component Restoration Models of Highway Bridges

Main Creators:

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OUTLINE

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BACKGROUND AND MOTIVATION

As one of the core topics of the "14th Five-Year Plan" and the 2035 Vision of National Development Strategy in China, the construction of "resilient cities" has become a major direction for disaster prevention and mitigation in China, with significant implications for the long-term stable development of the economy and society. Among natural disasters, earthquakes have been one of the most devastating hazards. Seismic resilience refers to the ability of a structure to maintain or restore its original functionality after being subjected to earthquakes. As a critical part of transportation infrastructure, the seismic design of bridge engineering is transitioning from the performance-based approach to a resilience-oriented concept. However, high-fidelity methods for resilience assessment are far from being fully developed, with the absence of post-earthquake functionality restoration models suitable for bridges in China. The restoration models are yet to be established due to the limited availability of field data on post-earthquake inspections and repairs in China. Instead, expert opinion surveys serve as a highly effective alternative, which is also the primary method used in the U.S. to develop restoration models of bridges (e.g., HAZUS earthquake models).

Component Restoration Model

This survey aims to collect experts' opinions on key variables of *component restoration models* of ordinary reinforced concrete highway bridges in China. The component restoration model is defined as a function that describes the functionality loss and recovery process of the bridge with damage to only this type of component after an earthquake. **Figure 1** shows the component restoration model, in which there are six key variables as listed in **Table 1**. It should be noted that the component restoration model is dependent on damage states of the component. Therefore, a comprehensive summary of damage states of the main components in highway bridges, including columns, bearings and decks, abutments and approach slabs, foundations, shear keys, and expansion joints, is given before answering the questions.

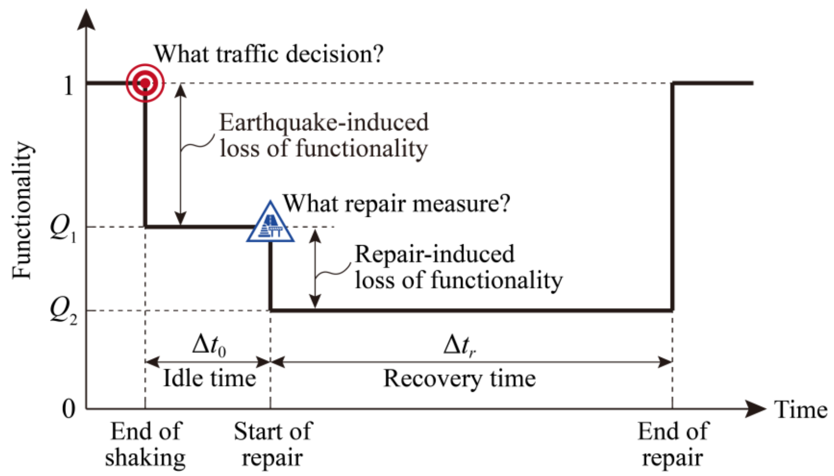

Figure 1. Schematic illustrations of component restoration models for bridges.

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Table 1. Key variables of proposed component restoration model.

#	Variables	Notation	Description
1	Earthquake-induced residual functionality	Q_1	The earthquake-induced loss of functionality for a bridge, assuming with a solely damaged type of component.
2	Traffic decision	—	The decision on the functionality level of a bridge, depending on the most severe damage state of the solely damaged type of component. Options for this decision include *fully open* (normal operation), *mostly open* (lane/speed restriction), *partially closed* (lane, speed and load restrictions), and *fully closed* (complete closure).
3	Idle time	Δt_0	The total time for post-earthquake reconnaissance, decision- and plan-making of repair strategies and measures, and preparation of crews, construction materials, equipment, etc.
4	Repair measure	—	The common repair measure for the damaged type of component.
5	Recovery time	Δt_r	The time required to repair the damaged type of component to as-built condition.
6	Repair-induced residual functionality	Q_2	The further reduced residual functionality induced by the repair of the damaged component.

More specifically, functionality, Q is represented by a normalized variable that $Q = 1$ refers to the full functionality, i.e., fully open, while $Q = 0$ indicates a complete loss of functionality, i.e., fully closure. As seen in **Figure 1**, after an earthquake event, stakeholders of a bridge are concerned with how much functionality is left, as defined by *earthquake-induced residual functionality*, Q_1, and what *traffic decision* should be made, as detailed in **Table 2**. Then, there is a period of *idle time*, Δt_0 between the occurrence of the earthquake and the start of any repair work. This period may involve but not limited to the time required for post-earthquake reconnaissance, decision-making of repair strategies, and preparation of all the repair resources including crews, materials, equipment, etc. Note that the idle time does not include any repair or construction work on the bridge or delays due to any other co-seismic hazards. Hence, the functionality is reasonably assumed invariant during the idle period, resulting in the flat line for this period in the restoration model.

Next, the repair process takes place, which may trigger additional loss of functionality, leading to a proposed key variable called *repair-induced residual functionality*, Q_2. For better understanding, suppose a scenario that a bridge experienced apparent bearing deformation after an earthquake, resulting in an obvious displacement of the deck (but yet to unseat). Accordingly, the bridge is perhaps partially open with speed restriction and an amount of loss of functionality can be estimated (i.e., $1-Q_1$). A common repair process for such a scenario is to raise the deck, replace the damaged bearings, and finally restore the deck. This repair process inevitably causes an additional loss of functionality (i.e., Q_1-Q_2) and keeps consistent (i.e., a flat line in **Figure 1**) across the *recovery time*, Δt_r until the full restoration of the damaged component.

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Basic Assumptions

To develop nationwide generalized component restoration models, please particularly be aware of the following four assumptions when you answer the following questions.

- **Assumption 1:** The bridge is fully functional before earthquake and accessible after the earthquake without considering traffic restriction factors for any areas.
- **Assumption 2:** When estimating the idle time, Δt_0, the required all resources for restoration including but not limited to crews, materials, equipment, etc. can be readily obtained as usual without any unexpectable delay.
- **Assumption 3:** Q_1 is estimated by the damage state of component, while Q_2 and Δt_r are estimated according to prevalent repair measures for the damaged components in China. Cutting-edge technologies yet to be used widely in practice are not considered.
- **Assumption 4:** The required crews, materials, and equipment resources for the repair of a single bridge is adequate, such that the same type of components (e.g., two columns in a bent) can be restored in parallel. This has been a common routine in China according to recent post-earthquake bridge repair experiences.

PERSONAL INFORMATION

Name: ________________________

Email or phone: _____________________

Affiliation: ______________________

Area of affiliation: ______________________ (Please choose one from below)

A. Central (华中) B. East (华东) C. North (华北) D. Northeast (东北)
E. Northwest (西北中) F. South (华南) G. Southwest (西南)

Professional title: _____________________ (Please choose one from below)

A. Senior (正高级) B. Associate Senior (副高级) C. Middle (中级) D. Junior (初级)

Career length: _____________________ (Please input an integer number)

Do you have post-earthquake inspection experience? _______________ (Please choose Yes or No)

DEFINITIONS OF COMPONENT DAMAGE STATES

As summarized in **Table 2**, five damage states are defined for most components except for the expansion joint, namely (almost) intact, slight, moderate, severe, and complete damage, as denoted by DS-1 to DS-5. As for the expansion joint, two limit states of intact (DS-1) and damaged (DS-2) are defined, because repair practices commonly replace the expansion joint as long as it is damaged.

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Table 2. Quantitative thresholds for multi-level damage states of bridge components.

		Damage states and quantitative thresholds				
		DS-1	DS-2	DS-3	DS-4	DS-5
Component	Engineering demand parameter (unit, if applicable)	(almost) Intact	Slight	Moderate	Severe	Complete
ULRB and FSB[1]	Residual displacement	0	$(0, s]$	$(s, s+D/4]$	$(s+D/4, s+D/2]$	$>s+D/2$
	Reduction percentage of carrying area	0	(0, 12.5%]	(12.5%, 25%]	(25%, 50%]	>50%
Rubber deformation-based isolation bearing[2]	Residual displacement	0	$(0, 0.5\cdot\Sigma t]$	$(0.5\cdot\Sigma t, 1.0\cdot\Sigma t]$	$(1.0\cdot\Sigma t, 2.0\cdot\Sigma t]$	$>2.0\cdot\Sigma t$
	Peak displacement	$(0, 1.75\cdot\Sigma t]$	$(1.75\cdot\Sigma t, 2.5\cdot\Sigma t]$	$(2.5\cdot\Sigma t, 3.0\cdot\Sigma t]$	$(3.0\cdot\Sigma t, 3.5\cdot\Sigma t]$	$>3.5\cdot\Sigma t$
	Reduction percentage of carrying area	0	(0, 15%]	(0, 25%]	(25%, 50%]	>50%
Fixed bearing	Reduction percentage of carrying area	0	(0, 12.5%]	(12.5%, 25%]	(25%, 50%]	>50%
Column	Peak drift ratio	[0, 1.9%]	(1.9%, 3.0%]	(3.0%, 4.3%]	(4.3%, 5.6%]	>5.6%
	Residual drift ratio	[0, 0.1%]	(0.1%, 0.2%]	(0.2%, 0.3%]	(0.3%, 0.7%]	>0.7%
Pile-group foundation	Peak displacement of cap (mm)	(0, 20]	(20, 35]	(35, 50]	(50, 70]	>70
	Residual displacement of cap (mm)	0	(0, 14]	(14, 25]	(25, 40]	>40
	Residual rotation of cap (rad)	0	(0, 0.004]	(0.004, 0.007]	(0.007, 0.01]	>0.01
Abutment and approach slab	Settlement of step-offset approach slab (mm)	(0, 1.27]	(1.27, 6.35]	(6.35, 17.78]	(17.78, 28.7]	>28.7
	Settlement of slope-offset approach slab (mm)	(0, 8.13]	(8.13, 16.26]	(16.26, 32.26]	(32.26, 64.47]	>64.47
Shear key	Displacement at shear key top (mm)	(0, 4]	(4, 17.6]	(17.6, 63.2]	(63.2, 105.3]	>105.3
Expansion joint[3]	Admissible displacement	Within	Exceeding	—	—	—

[1] Unbonded laminated rubber bearing (ULRB) and friction sliding bearing (FSB) are both friction sliding-dominated bearings. Definitions of s and D in residual displacement thresholds refer to **Figure 2** for ULRB and **Figure 3** for two common types of friction sliding bearings (pot and spherical). Note that scenarios with bearing peak displacements exceeding $s+D$ are taken as DS-5, while DS-1 to DS-4 shall meet an additional criterion that the bearing peak displacements do not exceed $s+D$.

[2] Σt in residual and peak displacement thresholds indicates the total thickness of laminated rubber.

[3] Damaged expansion joints are often replaced, thereby only two states are considered, i.e., within or exceeding the admissible displacement. The threshold of the admissible displacement can be determined as per the instruction of the specific expansion joint device.

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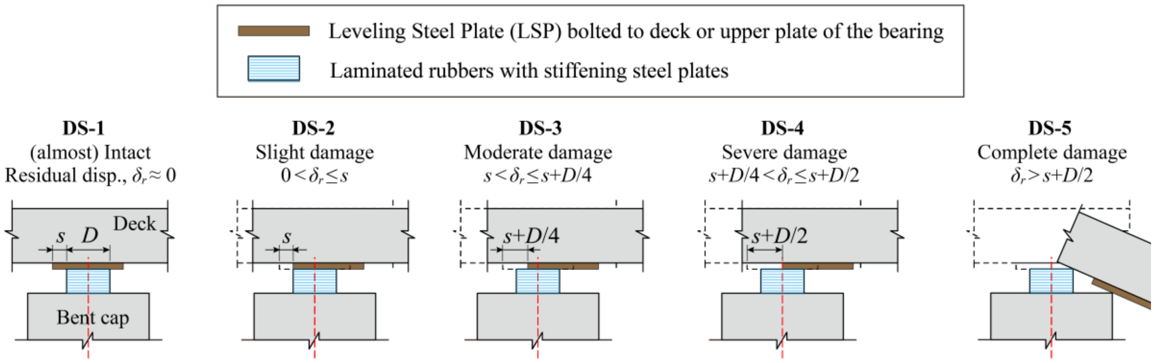

Figure 2. Illustration of multi-level damage states (DS-1 to DS-5) for unbonded laminated rubber bearings (ULRBs). (Note: plots of DS-1 to DS-4 display upper thresholds of the damage states.)

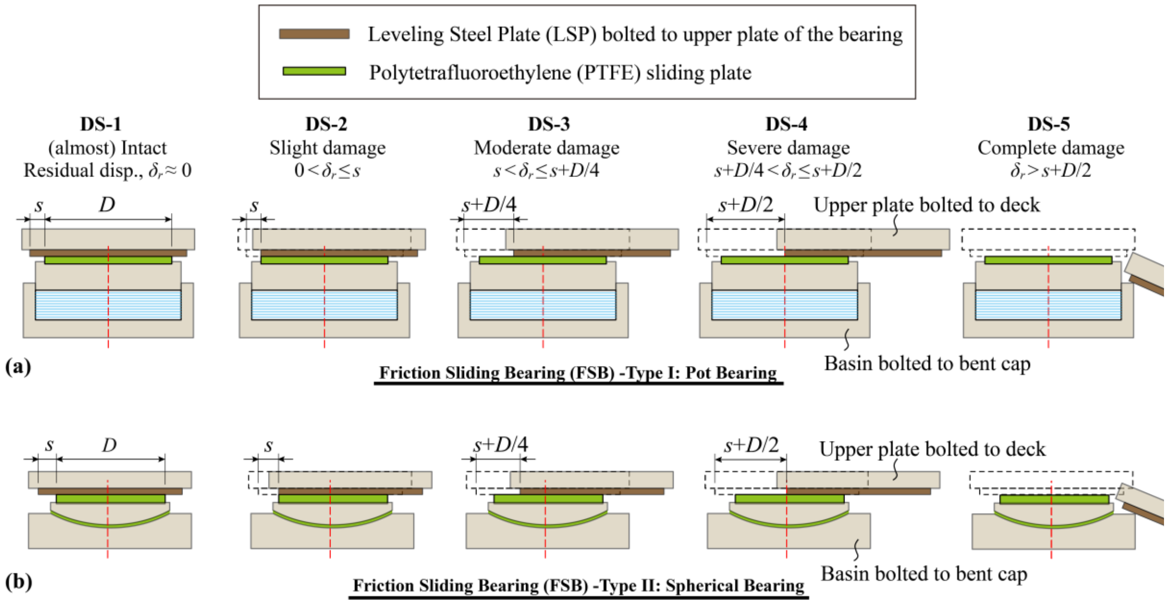

Figure 3. Illustration of multi-level damage states (DS-1 to DS-5) for friction sliding bearings (FSBs): (a) Type I: Pot bearing, and (b) Type II: Spherical bearing. (Note: plots of DS-1 to DS-4 display upper thresholds of the damage states.)

SURVEY OF KEY VARIABLES IN RESTORATION MODELS

Tables 3 to **10** display questionnaires for surveying the six key variables in restoration models of different components. Please choose one answer from the provided options for each question, and input remarks in the rightmost column of the table, if applicable.

Attention: Please be aware of the type of component from the table caption and the damage state from the leftmost column of each table.

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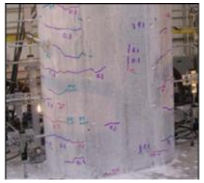
RC Column

Table 3. Survey of **RC Column** restoration model variables.

Damage state	Qualitative damage description	Quantitative damage indices	(Please choose one answer from the provided options for each question)						
			Traffic decision	Q_1 (%)	Δt_0 (day)	Most likely repair measure	Q_2 (%)	Δt_r (day)	Remarks
(almost) Intact (DS-1)	Few very minor horizontal cracks, no any other visible damage, e.g., 	γ_p: [0, 1.9%], or γ_r: [0, 0.1%], whichever occurs first. Note: γ_p: peak drift ratio; γ_r: residual drift ratio	A. Fully open (normal operation) B. Mostly open (lane or speed restriction) C. Partially close (lane, speed and load restrictions) D. Fully close (complete closure)	A. 0 B. 0-20 C. 20-40 D. 40-60 E. 60-80 F. 80-100 G. 100	A. 0 B. 0-7 C. 7-14 D. 14-21 E. 21-30 F. 30-60 G. 60-90 H. 90-180	A. No need to repair B. Concrete surface sealing, filling, etc. C. Repaired by jacketing with steel plate, fiber reinforced plastic straps, etc. D. Repaired by additional reinforcement and concreting E. Demolish and rebuild F. Other (please detail in the last column as remarks)	A. 0 B. 0-20 C. 20-40 D. 40-60 E. 60-80 F. 80-100	A. 0 B. 0-7 C. 7-14 D. 14-30 E. 30-60 F. 60-90 G. 90-180 H. 180-360	
Slight (DS-2)	First spalling of concrete, and possible minor diagonal cracks, e.g., 	γ_p: (1.9%, 3.0%], or γ_r: (0.1%, 0.2%], whichever occurs first. Note: γ_p: peak drift ratio; γ_r: residual drift ratio	A. Fully open (normal operation) B. Mostly open (lane or speed restriction) C. Partially close (lane, speed and load restrictions) D. Fully close (complete closure)	A. 0 B. 0-20 C. 20-40 D. 40-60 E. 60-80 F. 80-100 G. 100	A. 0 B. 0-7 C. 7-14 D. 14-21 E. 21-30 F. 30-60 G. 60-90 H. 90-180	A. No need to repair B. Concrete surface sealing, filling, etc. C. Repaired by jacketing with steel plate, fiber reinforced plastic straps, etc. D. Repaired by additional reinforcement and concreting E. Demolish and rebuild F. Other (please detail in the last column as remarks)	A. 0 B. 0-20 C. 20-40 D. 40-60 E. 60-80 F. 80-100	A. 0 B. 0-7 C. 7-14 D. 14-30 E. 30-60 F. 60-90 G. 90-180 H. 180-360	

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Damage state	Qualitative damage description	Quantitative damage indices	(Please choose one answer from the provided options for each question)						
			Traffic decision	Q_1 (%)	Δt_0 (day)	Most likely repair measure	Q_2 (%)	Δt_r (day)	Remarks
Moderate (DS-3)	More wide and deep cracks, and large amount of cover spalled, e.g., 	γ_p: (3.0%, 4.3%], or γ_r: (0.2%, 0.3%], whichever occurs first. Note: γ_p: peak drift ratio; γ_r: residual drift ratio	A. Fully open (normal operation) B. Mostly open (lane or speed restriction) C. Partially close (lane, speed and load restrictions) D. Fully close (complete closure)	A. 0 B. 0-20 C. 20-40 D. 40-60 E. 60-80 F. 80-100 G. 100	A. 0 B. 0-7 C. 7-14 D. 14-21 E. 21-30 F. 30-60 G. 60-90 H. 90-180	A. No need to repair B. Concrete surface sealing, filling, etc. C. Repaired by jacketing with steel plate, fiber reinforced plastic straps, etc. D. Repaired by additional reinforcement and concreting E. Demolish and rebuild F. Other (please detail in the last column as remarks)	A. 0 B. 0-20 C. 20-40 D. 40-60 E. 60-80 F. 80-100	A. 0 B. 0-7 C. 7-14 D. 14-30 E. 30-60 F. 60-90 G. 90-180 H. 180-360	
Severe (DS-4)	Visible rebars and confining bars, e.g., 	γ_p: (4.3%, 5.6%], or γ_r: (0.3%, 0.7%], whichever occurs first. Note: γ_p: peak drift ratio; γ_r: residual drift ratio	A. Fully open (normal operation) B. Mostly open (lane or speed restriction) C. Partially close (lane, speed and load restrictions) D. Fully close (complete closure)	A. 0 B. 0-20 C. 20-40 D. 40-60 E. 60-80 F. 80-100 G. 100	A. 0 B. 0-7 C. 7-14 D. 14-21 E. 21-30 F. 30-60 G. 60-90 H. 90-180	A. No need to repair B. Concrete surface sealing, filling, etc. C. Repaired by jacketing with steel plate, fiber reinforced plastic straps, etc. D. Repaired by additional reinforcement and concreting E. Demolish and rebuild F. Other (please detail in the last column as remarks)	A. 0 B. 0-20 C. 20-40 D. 40-60 E. 60-80 F. 80-100	A. 0 B. 0-7 C. 7-14 D. 14-30 E. 30-60 F. 60-90 G. 90-180 H. 180-360	

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Damage state	Qualitative damage description	Quantitative damage indices	(Please choose one answer from the provided options for each question)						
			Traffic decision	Q_1 (%)	Δt_0 (day)	Most likely repair measure	Q_2 (%)	Δt_r (day)	Remarks
Complete (DS-5)	Rebar buckling or concrete core crushing, e.g., 	γ_p: >5.6%, or γ_r: >0.7%, whichever occurs first. Note: γ_p: peak drift ratio; γ_r: residual drift ratio	A. Fully open (normal operation) B. Mostly open (lane or speed restriction) C. Partially close (lane, speed and load restrictions) D. Fully close (complete closure)	A. 0 B. 0-20 C. 20-40 D. 40-60 E. 60-80 F. 80-100 G. 100	A. 0 B. 0-7 C. 7-14 D. 14-21 E. 21-30 F. 30-60 G. 60-90 H. 90-180	A. No need to repair B. Concrete surface sealing, filling, etc. C. Repaired by jacketing with steel plate, fiber reinforced plastic straps, etc. D. Repaired by additional reinforcement and concreting E. Demolish and rebuild F. Other (please detail in the last column as remarks)	A. 0 B. 0-20 C. 20-40 D. 40-60 E. 60-80 F. 80-100	A. 0 B. 0-7 C. 7-14 D. 14-30 E. 30-60 F. 60-90 G. 90-180 H. 180-360	

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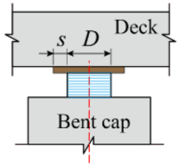
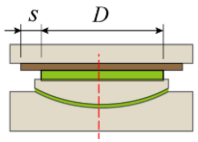
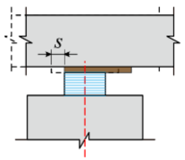
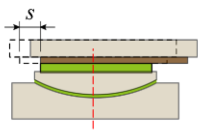
ULRB and FSB (with Deck)

Table 4. Survey of Unbonded Laminated Rubber Bearing **(ULRB)** and Friction Sliding Bearing **(FSB) (with Deck)** restoration model variables.

Damage state	Qualitative damage description	Quantitative damage indices	(Please choose one answer from the provided options for each question)						
			Traffic decision	Q_1 (%)	Δt_0 (day)	Most likely repair measure	Q_2 (%)	Δt_r (day)	Remarks
(almost) Intact (DS-1)	No residual displacement, e.g.,  (ULRB)  (FSB-Spherical)	$\delta_r = 0$ and $R_A = 0$, both satisfied. Note: δ_r: residual bearing displacement R_A: reduction percentage of carrying area	A. Fully open (normal operation) B. Mostly open (lane or speed restriction) C. Partially close (lane, speed and load restrictions) D. Fully close (complete closure)	A. 0 B. 0-20 C. 20-40 D. 40-60 E. 60-80 F. 80-100 G. 100	A. 0 B. 0-7 C. 7-14 D. 14-21 E. 21-30 F. 30-60 G. 60-90 H. 90-180	A. No need to repair B. Restore the deck only C. Replace the bearing and then restore the deck D. Other (please detail in the last column as remarks)	A. 0 B. 0-20 C. 20-40 D. 40-60 E. 60-80 F. 80-100	A. 0 B. 0-7 C. 7-14 D. 14-30 E. 30-60 F. 60-90 G. 90-180 H. 180-360	
Slight (DS-2)	Slight residual displacement; full carrying area remains, i.e., upper limits:  (ULRB)  (FSB-Spherical)	δ_r: (0, s] or R_A: (0, 12.5%], whichever occurs first. Note: δ_r: residual bearing displacement R_A: reduction percentage of carrying area	A. Fully open (normal operation) B. Mostly open (lane or speed restriction) C. Partially close (lane, speed and load restrictions) D. Fully close (complete closure)	A. 0 B. 0-20 C. 20-40 D. 40-60 E. 60-80 F. 80-100 G. 100	A. 0 B. 0-7 C. 7-14 D. 14-21 E. 21-30 F. 30-60 G. 60-90 H. 90-180	A. No need to repair B. Restore the deck only C. Replace the bearing and then restore the deck D. Other (please detail in the last column as remarks)	A. 0 B. 0-20 C. 20-40 D. 40-60 E. 60-80 F. 80-100	A. 0 B. 0-7 C. 7-14 D. 14-30 E. 30-60 F. 60-90 G. 90-180 H. 180-360	

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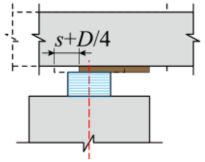
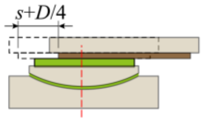
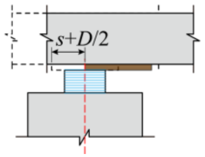
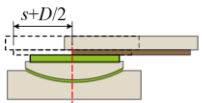
Damage state	Qualitative damage description	Quantitative damage indices	(Please choose one answer from the provided options for each question)						
			Traffic decision	Q_1 (%)	Δt_0 (day)	Most likely repair measure	Q_2 (%)	Δt_r (day)	Remarks
Moderate (DS-3)	Moderate residual displacement; but still with considerable carrying area, i.e., upper limits:  (ULRB)  (FSB-Spherical)	δ_r: $(s, s+D/4]$ or R_A: (12.5%, 25%], whichever occurs first. Note: δ_r: residual bearing displacement R_A: reduction percentage of carrying area	A. Fully open (normal operation) B. Mostly open (lane or speed restriction) C. Partially close (lane, speed and load restrictions) D. Fully close (complete closure)	A. 0 B. 0-20 C. 20-40 D. 40-60 E. 60-80 F. 80-100 G. 100	I. 0 A. 0-7 B. 7-14 C. 14-21 D. 21-30 E. 30-60 F. 60-90 G. 90-180	A. No need to repair B. Restore the deck only C. Replace the bearing and then restore the deck D. Other (please detail in the last column as remarks)	A. 0 B. 0-20 C. 20-40 D. 40-60 E. 60-80 F. 80-100	A. 0 B. 0-7 C. 7-14 D. 14-30 E. 30-60 F. 60-90 G. 90-180 H. 180-360	
Severe (DS-4)	Large residual displacement, upper limits:  (ULRB)  (FSB-Spherical)	δ_r: $(s+D/4, s+D/2]$ or R_A: (25%, 50%], whichever occurs first. Note: δ_r: residual bearing displacement R_A: reduction percentage of carrying area	A. Fully open (normal operation) B. Mostly open (lane or speed restriction) C. Partially close (lane, speed and load restrictions) D. Fully close (complete closure)	A. 0 B. 0-20 C. 20-40 D. 40-60 E. 60-80 F. 80-100 G. 100	A. 0 B. 0-7 C. 7-14 D. 14-21 E. 21-30 F. 30-60 G. 60-90 H. 90-180	A. No need to repair B. Restore the deck only C. Replace the bearing and then restore the deck D. Other (please detail in the last column as remarks)	A. 0 B. 0-20 C. 20-40 D. 40-60 E. 60-80 F. 80-100	A. 0 B. 0-7 C. 7-14 D. 14-30 E. 30-60 F. 60-90 G. 90-180 H. 180-360	

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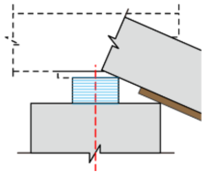
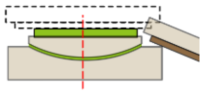
Damage state	Qualitative damage description	Quantitative damage indices	(Please choose one answer from the provided options for each question)						
			Traffic decision	Q_1 (%)	Δt_0 (day)	Most likely repair measure	Q_2 (%)	Δt_r (day)	Remarks
Complete (DS-5)	Very large residual displacement beyond half of the bearing width or diameter, or no effective carrying area, high risk to unseating or even falling of the deck, whichever occurs first  (ULRB)  (FSB-Spherical)	δ_r: $>s+D/2$ or R_A: >50%, whichever occurs first. Note: δ_r: residual bearing displacement R_A: reduction percentage of carrying area	A. Fully open (normal operation) B. Mostly open (lane or speed restriction) C. Partially close (lane, speed and load restrictions) D. Fully close (complete closure)	A. 0 B. 0-20 C. 20-40 D. 40-60 E. 60-80 F. 80-100 G. 100	A. 0 B. 0-7 C. 7-14 D. 14-21 E. 21-30 F. 30-60 G. 60-90 H. 90-180	A. No need to repair B. Restore the deck only C. Replace the bearing and then restore the deck D. Other (please detail in the last column as remarks)	A. 0 B. 0-20 C. 20-40 D. 40-60 E. 60-80 F. 80-100	A. 0 B. 0-7 C. 7-14 D. 14-30 E. 30-60 F. 60-90 G. 90-180 H. 180-360	

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Rubber Deformation-based Isolation Bearing (with Deck)

Table 5. Survey of **Rubber Deformation-based Isolation Bearing (with Deck)** restoration model variables.

Damage state	Qualitative damage description	Quantitative damage indices	(Please choose one answer from the provided options for each question)						
			Traffic decision	Q_1 (%)	Δt_0 (day)	Most likely repair measure	Q_2 (%)	Δt_r (day)	Remarks
(almost) Intact (DS-1)	No residual deformation with respect to the total thickness of the rubber, small peak deformation, and full carrying area remains. Note: Σt: total thickness of the rubbers	$\delta_r = 0$, δ_p: (0, 1.75·Σt], and R_A =0, all satisfied. Note: δ_r: residual bearing deformation δ_p: peak bearing deformation	A. Fully open (normal operation) B. Mostly open (lane or speed restriction) C. Partially close (lane, speed and load restrictions) D. Fully close (complete closure)	A. 0 B. 0-20 C. 20-40 D. 40-60 E. 60-80 F. 80-100 G. 100	A. 0 B. 0-7 C. 7-14 D. 14-21 E. 21-30 F. 30-60 G. 60-90 H. 90-180	A. No need to repair B. Restore the deck only C. Replace the bearing and then restore the deck D. Other (please detail in the last column as remarks)	A. 0 B. 0-20 C. 20-40 D. 40-60 E. 60-80 F. 80-100	A. 0 B. 0-7 C. 7-14 D. 14-30 E. 30-60 F. 60-90 G. 90-180 H. 180-360	
Slight (DS-2)	Slight residual deformation, slight peak deformation, and slight reduction of carrying area. Note: Σt: total thickness of the rubbers	δ_r: (0, 0.5·Σt], δ_p: (1.75·Σt, 2.5·Σt], or R_A: (0, 15%], whichever occurs first Note: δ_r: residual bearing deformation δ_p: peak bearing deformation R_A: reduction percentage of carrying area	A. Fully open (normal operation) B. Mostly open (lane or speed restriction) C. Partially close (lane, speed and load restrictions) D. Fully close (complete closure)	A. 0 B. 0-20 C. 20-40 D. 40-60 E. 60-80 F. 80-100 G. 100	A. 0 B. 0-7 C. 7-14 D. 14-21 E. 21-30 F. 30-60 G. 60-90 H. 90-180	A. No need to repair B. Restore the deck only C. Replace the bearing and then restore the deck D. Other (please detail in the last column as remarks)	A. 0 B. 0-20 C. 20-40 D. 40-60 E. 60-80 F. 80-100	A. 0 B. 0-7 C. 7-14 D. 14-30 E. 30-60 F. 60-90 G. 90-180 H. 180-360	

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Damage state	Qualitative damage description	Quantitative damage indices	(Please choose one answer from the provided options for each question)						
			Traffic decision	Q_1 (%)	Δt_0 (day)	Most likely repair measure	Q_2 (%)	Δt_r (day)	Remarks
Moderate (DS-3)	Moderate residual deformation, moderate peak deformation, and moderate reduction of carrying area. Note: Σt: total thickness of the rubbers	δ_r: $(0.5{\cdot}\Sigma t, 1.0{\cdot}\Sigma t]$, δ_p: $(2.5{\cdot}\Sigma t, 3.0{\cdot}\Sigma t]$, or R_A: (15%, 25%], whichever occurs first Note: δ_r: residual bearing deformation δ_p: peak bearing deformation R_A: reduction percentage of carrying area	A. Fully open (normal operation) B. Mostly open (lane or speed restriction) C. Partially close (lane, speed and load restrictions) D. Fully close (complete closure)	A. 0 B. 0-20 C. 20-40 D. 40-60 E. 60-80 F. 80-100 G. 100	A. 0 B. 0-7 C. 7-14 D. 14-21 E. 21-30 F. 30-60 G. 60-90 H. 90-180	A. No need to repair B. Restore the deck only C. Replace the bearing and then restore the deck D. Other (please detail in the last column as remarks)	A. 0 B. 0-20 C. 20-40 D. 40-60 E. 60-80 F. 80-100	A. 0 B. 0-7 C. 7-14 D. 14-30 E. 30-60 F. 60-90 G. 90-180 H. 180-360	
Severe (DS-4)	Considerable residual deformation, peak deformation, and reduction of carrying area. Note: Σt: total thickness of the rubbers	δ_r: $(1.0{\cdot}\Sigma t, 2.0{\cdot}\Sigma t]$, δ_p: $(3.0{\cdot}\Sigma t, 3.5{\cdot}\Sigma t]$, or R_A: (25%, 50%], whichever occurs first Note: δ_r: residual bearing deformation δ_p: peak bearing deformation R_A: reduction percentage of carrying area	A. Fully open (normal operation) B. Mostly open (lane or speed restriction) C. Partially close (lane, speed and load restrictions) D. Fully close (complete closure)	A. 0 B. 0-20 C. 20-40 D. 40-60 E. 60-80 F. 80-100 G. 100	A. 0 B. 0-7 C. 7-14 D. 14-21 E. 21-30 F. 30-60 G. 60-90 H. 90-180	A. No need to repair B. Restore the deck only C. Replace the bearing and then restore the deck D. Other (please detail in the last column as remarks)	A. 0 B. 0-20 C. 20-40 D. 40-60 E. 60-80 F. 80-100	A. 0 B. 0-7 C. 7-14 D. 14-30 E. 30-60 F. 60-90 G. 90-180 H. 180-360	

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Damage state	Qualitative damage description	Quantitative damage indices	(Please choose one answer from the provided options for each question)						
			Traffic decision	Q_1 (%)	Δt_0 (day)	Most likely repair measure	Q_2 (%)	Δt_r (day)	Remarks
Complete (DS-5)	Very large residual deformation, peak deformation, and reduction of carrying area. Note: Σt: total thickness of the rubbers	δ_r: >2.0·Σt, δ_p: >3.5·Σt, or R_A: >50%, whichever occurs first Note: δ_r: residual bearing deformation δ_p: peak bearing deformation R_A: reduction percentage of carrying area	A. Fully open (normal operation) B. Mostly open (lane or speed restriction) C. Partially close (lane, speed and load restrictions) D. Fully close (complete closure)	A. 0 B. 0-20 C. 20-40 D. 40-60 E. 60-80 F. 80-100 G. 100	A. 0 B. 0-7 C. 7-14 D. 14-21 E. 21-30 F. 30-60 G. 60-90 H. 90-180	A. No need to repair B. Restore the deck only C. Replace the bearing and then restore the deck D. Other (please detail in the last column as remarks)	A. 0 B. 0-20 C. 20-40 D. 40-60 E. 60-80 F. 80-100	A. 0 B. 0-7 C. 7-14 D. 14-30 E. 30-60 F. 60-90 G. 90-180 H. 180-360	

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Fixed Bearing (with Deck)

Table 6. Survey of **Fixed Bearing (with Deck)** restoration model variables.

Damage state	Qualitative damage description	Quantitative damage indices	(Please choose one answer from the provided options for each question)						
			Traffic decision	Q_1 (%)	Δt_0 (day)	Most likely repair measure	Q_2 (%)	Δt_r (day)	Remarks
(almost) Intact (DS-1)	No any visible displacement or damage.	R_A =0 Note: R_A: reduction percentage of carrying area	A. Fully open (normal operation) B. Mostly open (lane or speed restriction) C. Partially close (lane, speed and load restrictions) D. Fully close (complete closure)	A. 0 B. 0-20 C. 20-40 D. 40-60 E. 60-80 F. 80-100 G. 100	A. 0 B. 0-7 C. 7-14 D. 14-21 E. 21-30 F. 30-60 G. 60-90 H. 90-180	A. No need to repair B. Restore the deck only C. Replace the bearing and then restore the deck D. Other (please detail in the last column as remarks)	A. 0 B. 0-20 C. 20-40 D. 40-60 E. 60-80 F. 80-100	A. 0 B. 0-7 C. 7-14 D. 14-30 E. 30-60 F. 60-90 G. 90-180 H. 180-360	
Slight (DS-2)	Damage to the limit basin ring, but very slight displacement and slight reduction of carrying area	R_A: (0, 15%] Note: R_A: reduction percentage of carrying area	A. Fully open (normal operation) B. Mostly open (lane or speed restriction) C. Partially close (lane, speed and load restrictions) D. Fully close (complete closure)	A. 0 B. 0-20 C. 20-40 D. 40-60 E. 60-80 F. 80-100 G. 100	A. 0 B. 0-7 C. 7-14 D. 14-21 E. 21-30 F. 30-60 G. 60-90 H. 90-180	A. No need to repair B. Restore the deck only C. Replace the bearing and then restore the deck D. Other (please detail in the last column as remarks)	A. 0 B. 0-20 C. 20-40 D. 40-60 E. 60-80 F. 80-100	A. 0 B. 0-7 C. 7-14 D. 14-30 E. 30-60 F. 60-90 G. 90-180 H. 180-360	

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Damage state	Qualitative damage description	Quantitative damage indices	(Please choose one answer from the provided options for each question)						
			Traffic decision	Q_1 (%)	Δt_0 (day)	Most likely repair measure	Q_2 (%)	Δt_r (day)	Remarks
Moderate (DS-3)	Obvious moderate displacement, and reduction of carrying area	R_A: (15%, 25%], Note: R_A: reduction percentage of carrying area	A. Fully open (normal operation) B. Mostly open (lane or speed restriction) C. Partially close (lane, speed and load restrictions) D. Fully close (complete closure)	A. 0 B. 0-20 C. 20-40 D. 40-60 E. 60-80 F. 80-100 G. 100	A. 0 B. 0-7 C. 7-14 D. 14-21 E. 21-30 F. 30-60 G. 60-90 H. 90-180	A. No need to repair B. Restore the deck only C. Replace the bearing and then restore the deck D. Other (please detail in the last column as remarks)	A. 0 B. 0-20 C. 20-40 D. 40-60 E. 60-80 F. 80-100	A. 0 B. 0-7 C. 7-14 D. 14-30 E. 30-60 F. 60-90 G. 90-180 H. 180-360	
Severe (DS-4)	Severe displacement, even with partial separation and considerable reduction of carrying area	R_A: (25%, 50%], Note: R_A: reduction percentage of carrying area	A. Fully open (normal operation) B. Mostly open (lane or speed restriction) C. Partially close (lane, speed and load restrictions) D. Fully close (complete closure)	A. 0 B. 0-20 C. 20-40 D. 40-60 E. 60-80 F. 80-100 G. 100	A. 0 B. 0-7 C. 7-14 D. 14-21 E. 21-30 F. 30-60 G. 60-90 H. 90-180	A. No need to repair B. Restore the deck only C. Replace the bearing and then restore the deck D. Other (please detail in the last column as remarks)	A. 0 B. 0-20 C. 20-40 D. 40-60 E. 60-80 F. 80-100	A. 0 B. 0-7 C. 7-14 D. 14-30 E. 30-60 F. 60-90 G. 90-180 H. 180-360	

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Damage state	Qualitative damage description	Quantitative damage indices	(Please choose one answer from the provided options for each question)						
			Traffic decision	Q_1 (%)	Δt_0 (day)	Most likely repair measure	Q_2 (%)	Δt_r (day)	Remarks
Complete (DS-5)	Completely damaged, and large loss of carrying area, with high risk of unseating or even falling of the deck	R_A: >50% Note: R_A: reduction percentage of carrying area	A. Fully open (normal operation) B. Mostly open (lane or speed restriction) C. Partially close (lane, speed and load restrictions) D. Fully close (complete closure)	A. 0 B. 0-20 C. 20-40 D. 40-60 E. 60-80 F. 80-100 G. 100	A. 0 B. 0-7 C. 7-14 D. 14-21 E. 21-30 F. 30-60 G. 60-90 H. 90-180	A. No need to repair B. Restore the deck only C. Replace the bearing and then restore the deck D. Other (please detail in the last column as remarks)	A. 0 B. 0-20 C. 20-40 D. 40-60 E. 60-80 F. 80-100	A. 0 B. 0-7 C. 7-14 D. 14-30 E. 30-60 F. 60-90 G. 90-180 H. 180-360	

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Pile-Group Foundation

Table 7. Survey of **Pile-Group Foundation** restoration model variables.

Damage state	Qualitative damage description	Quantitative damage indices	(Please choose one answer from the provided options for each question)						
			Traffic decision	Q_1 (%)	Δt_0 (day)	Most likely repair measure	Q_2 (%)	Δt_r (day)	Remarks
(almost) Intact (DS-1)	No visible displacement or rotation on the cap. The entire foundation should be elastic state	$\delta_{c,p}$: [0, 20], $\delta_{c,r}$ = 0 and θ_r = 0, all satisfied Note: $\delta_{c,p}$: peak cap displacement (mm) $\delta_{c,r}$: residual cap displacement (mm) θ_r: residual cap rotation (rad)	A. Fully open (normal operation) B. Mostly open (lane or speed restriction) C. Partially close (lane, speed and load restrictions) D. Fully close (complete closure)	A. 0 B. 0-20 C. 20-40 D. 40-60 E. 60-80 F. 80-100 G. 100	A. 0 B. 0-7 C. 7-14 D. 14-21 E. 21-30 F. 30-60 G. 60-90 H. 90-180	A. No need to repair B. Surface sealing, filling, etc. C. Repaired by jacketing with steel plate, fiber reinforced plastic straps, etc. D. Add piles to enhance the support E. Rebuild F. Other (please detail in the last column as remarks)	A. 0 B. 0-20 C. 20-40 D. 40-60 E. 60-80 F. 80-100	A. 0 B. 0-7 C. 7-14 D. 14-30 E. 30-60 F. 60-90 G. 90-180 H. 180-360	
Slight (DS-2)	Minor horizontal cracks on the pile head.	$\delta_{c,p}$: (20, 35], $\delta_{c,r}$: (0, 14] or θ_r: (0, 0.004], whichever occurs first Note: $\delta_{c,p}$: peak cap displacement (mm) $\delta_{c,r}$: residual cap displacement (mm) θ_r: residual cap rotation (rad)	A. Fully open (normal operation) B. Mostly open (lane or speed restriction) C. Partially close (lane, speed and load restrictions) E. Fully close (complete closure)	A. 0 B. 0-20 C. 20-40 D. 40-60 E. 60-80 F. 80-100 G. 100	A. 0 B. 0-7 C. 7-14 D. 14-21 E. 21-30 F. 30-60 G. 60-90 H. 90-180	A. No need to repair B. Surface sealing, filling, etc. C. Repaired by jacketing with steel plate, fiber reinforced plastic straps, etc. D. Add piles to enhance the support E. Rebuild F. Other (please detail in the last column as remarks)	A. 0 B. 0-20 C. 20-40 D. 40-60 E. 60-80 F. 80-100	A. 0 B. 0-7 C. 7-14 D. 14-30 E. 30-60 F. 60-90 G. 90-180 H. 180-360	

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Damage state	Qualitative damage description	Quantitative damage indices	(Please choose one answer from the provided options for each question)						
			Traffic decision	Q_1 (%)	Δt_0 (day)	Most likely repair measure	Q_2 (%)	Δt_r (day)	Remarks
Moderate (DS-3)	Concrete cover spalling at pile head, and occurrence of underground second plastic hinge.	$\delta_{c,p}$: (35, 50], $\delta_{c,r}$: (14, 25] or θ_r: (0.004, 0.007], whichever occurs first Note: $\delta_{c,p}$: peak cap displacement (mm) $\delta_{c,r}$: residual cap displacement (mm) θ_r: residual cap rotation (rad)	A. Fully open (normal operation) B. Mostly open (lane or speed restriction) C. Partially close (lane, speed and load restrictions) D. Fully close (complete closure)	A. 0 B. 0-20 C. 20-40 D. 40-60 E. 60-80 F. 80-100 G. 100	A. 0 B. 0-7 C. 7-14 D. 14-21 E. 21-30 F. 30-60 G. 60-90 H. 90-180	A. No need to repair B. Surface sealing, filling, etc. C. Repaired by jacketing with steel plate, fiber reinforced plastic straps, etc. D. Add piles to enhance the support E. Rebuild F. Other (please detail in the last column as remarks)	A. 0 B. 0-20 C. 20-40 D. 40-60 E. 60-80 F. 80-100	A. 0 B. 0-7 C. 7-14 D. 14-30 E. 30-60 F. 60-90 G. 90-180 H. 180-360	
Severe (DS-4)	Concrete cover extensive spalling at pile head. Underground second plastic hinge aggravates.	$\delta_{c,p}$: (50, 70], $\delta_{c,r}$: (25, 40] or θ_r: (0.007, 0.01], whichever occurs first Note: $\delta_{c,p}$: peak cap displacement (mm) $\delta_{c,r}$: residual cap displacement (mm) θ_r: residual cap rotation (rad)	A. Fully open (normal operation) B. Mostly open (lane or speed restriction) C. Partially close (lane, speed and load restrictions) D. Fully close (complete closure)	A. 0 B. 0-20 C. 20-40 D. 40-60 E. 60-80 F. 80-100 G. 100	A. 0 B. 0-7 C. 7-14 D. 14-21 E. 21-30 F. 30-60 G. 60-90 H. 90-180	A. No need to repair B. Surface sealing, filling, etc. C. Repaired by jacketing with steel plate, fiber reinforced plastic straps, etc. D. Add piles to enhance the support E. Rebuild F. Other (please detail in the last column as remarks)	A. 0 B. 0-20 C. 20-40 D. 40-60 E. 60-80 F. 80-100	A. 0 B. 0-7 C. 7-14 D. 14-30 E. 30-60 F. 60-90 G. 90-180 H. 180-360	

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Damage state	Qualitative damage description	Quantitative damage indices	(Please choose one answer from the provided options for each question)						
			Traffic decision	Q_1 (%)	Δt_0 (day)	Most likely repair measure	Q_2 (%)	Δt_r (day)	Remarks
Complete (DS-5)	Concrete core crushing or rebar snapping at pile head.	$\delta_{c,p}$: >70, $\delta_{c,r}$: >40 or θ_r: >0.01, whichever occurs first Note: $\delta_{c,p}$: peak cap displacement (mm) $\delta_{c,r}$: residual cap displacement (mm) θ_r: residual cap rotation (rad)	A. Fully open (normal operation) B. Mostly open (lane or speed restriction) C. Partially close (lane, speed and load restrictions) D. Fully close (complete closure)	A. 0 B. 0-20 C. 20-40 D. 40-60 E. 60-80 F. 80-100 G. 100	A. 0 B. 0-7 C. 7-14 D. 14-21 E. 21-30 F. 30-60 G. 60-90 H. 90-180	A. No need to repair B. Surface sealing, filling, etc. C. Repaired by jacketing with steel plate, fiber reinforced plastic straps, etc. D. Add piles to enhance the support E. Rebuild F. Other (please detail in the last column as remarks)	A. 0 B. 0-20 C. 20-40 D. 40-60 E. 60-80 F. 80-100	A. 0 B. 0-7 C. 7-14 D. 14-30 E. 30-60 F. 60-90 G. 90-180 H. 180-360	

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Abutment and Approach Slab

Table 8. Survey of **Abutment and Approach Slab** restoration model variables.

Damage state	Qualitative damage description	Quantitative damage indices	(Please choose one answer from the provided options for each question)						
			Traffic decision	Q_1 (%)	Δt_0 (day)	Most likely repair measure	Q_2 (%)	Δt_r (day)	Remarks
(almost) Intact (DS-1)	Very minor cracks on abutment stem, no any other visible damage on abutment. Very minor settlement of the approach slab.	δ_{step}: (0, 1.27], or δ_{slope}: (0, 8.13], whichever occurs first. Note: $\delta_{c,p}$: Settlement of step-offset approach slab (mm) $\delta_{c,r}$: Settlement of slope offset approach slab (mm)	A. Fully open (normal operation) B. Mostly open (lane or speed restriction) C. Partially close (lane, speed and load restrictions) D. Fully close (complete closure)	A. 0 B. 0-20 C. 20-40 D. 40-60 E. 60-80 F. 80-100 G. 100	A. 0 B. 0-7 C. 7-14 D. 14-21 E. 21-30 F. 30-60 G. 60-90 H. 90-180	A. No need to repair B. Concrete surface sealing, filling, etc. C. Repaired by jacketing with steel plate, fiber reinforced plastic straps, etc. D. Repaired by additional reinforcement and concreting E. Demolish and rebuild F. Other (please detail in the last column as remarks)	A. 0 B. 0-20 C. 20-40 D. 40-60 E. 60-80 F. 80-100	A. 0 B. 0-7 C. 7-14 D. 14-30 E. 30-60 F. 60-90 G. 90-180 H. 180-360	
Slight (DS-2)	Slight spalling of cover on abutment stem. Slight settlement of the approach slab.	δ_{step}: (0, 1.27], or δ_{slope}: (8.13, 16.26], whichever occurs first. Note: $\delta_{c,p}$: Settlement of step-offset approach slab (mm) $\delta_{c,r}$: Settlement of slope-offset approach slab (mm)	A. Fully open (normal operation) B. Mostly open (lane or speed restriction) C. Partially close (lane, speed and load restrictions) D. Fully close (complete closure)	A. 0 B. 0-20 C. 20-40 D. 40-60 E. 60-80 F. 80-100 G. 100	A. 0 B. 0-7 C. 7-14 D. 14-21 E. 21-30 F. 30-60 G. 60-90 H. 90-180	A. No need to repair B. Concrete surface sealing, filling, etc. C. Repaired by jacketing with steel plate, fiber reinforced plastic straps, etc. D. Repaired by additional reinforcement and concreting E. Demolish and rebuild F. Other (please detail in the last column as remarks)	A. 0 B. 0-20 C. 20-40 D. 40-60 E. 60-80 F. 80-100	A. 0 B. 0-7 C. 7-14 D. 14-30 E. 30-60 F. 60-90 G. 90-180 H. 180-360	

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Damage state	Qualitative damage description	Quantitative damage indices	(Please choose one answer from the provided options for each question)						
			Traffic decision	Q_1 (%)	Δt_0 (day)	Most likely repair measure	Q_2 (%)	Δt_r (day)	Remarks
Moderate (DS-3)	Concrete cover spalling aggravates at stem, but no rebar visible. Apparently moderate settlement of the approach slab.	δ_{step}: (6.35, 17.78], or δ_{slope}: (16.26, 32.26], whichever occurs first. Note: $\delta_{c,p}$: Settlement of step-offset approach slab (mm) $\delta_{c,r}$: Settlement of slope-offset approach slab (mm)	A. Fully open (normal operation) B. Mostly open (lane or speed restriction) C. Partially close (lane, speed and load restrictions) D. Fully close (complete closure)	A. 0 B. 0-20 C. 20-40 D. 40-60 E. 60-80 F. 80-100 G. 100	A. 0 B. 0-7 C. 7-14 D. 14-21 E. 21-30 F. 30-60 G. 60-90 H. 90-180	A. No need to repair B. Concrete surface sealing, filling, etc. C. Repaired by jacketing with steel plate, fiber reinforced plastic straps, etc. D. Repaired by additional reinforcement and concreting E. Demolish and rebuild F. Other (please detail in the last column as remarks)	A. 0 B. 0-20 C. 20-40 D. 40-60 E. 60-80 F. 80-100	A. 0 B. 0-7 C. 7-14 D. 14-30 E. 30-60 F. 60-90 G. 90-180 H. 180-360	
Severe (DS-4)	Concrete cover largely spalling at stem, and rebars are visible. Quite large settlement of the approach slab.	δ_{step}: (17.78, 28.7], or δ_{slope}: (32.26, 64.47], whichever occurs first. Note: $\delta_{c,p}$: Settlement of step-offset approach slab (mm) $\delta_{c,r}$: Settlement of slope-offset approach slab (mm)	A. Fully open (normal operation) B. Mostly open (lane or speed restriction) C. Partially close (lane, speed and load restrictions) D. Fully close (complete closure)	A. 0 B. 0-20 C. 20-40 D. 40-60 E. 60-80 F. 80-100 G. 100	A. 0 B. 0-7 C. 7-14 D. 14-21 E. 21-30 F. 30-60 G. 60-90 H. 90-180	A. No need to repair B. Concrete surface sealing, filling, etc. C. Repaired by jacketing with steel plate, fiber reinforced plastic straps, etc. D. Repaired by additional reinforcement and concreting E. Demolish and rebuild F. Other (please detail in the last column as remarks)	A. 0 B. 0-20 C. 20-40 D. 40-60 E. 60-80 F. 80-100	A. 0 B. 0-7 C. 7-14 D. 14-30 E. 30-60 F. 60-90 G. 90-180 H. 180-360	

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Damage state	Qualitative damage description	Quantitative damage indices	(Please choose one answer from the provided options for each question)						
			Traffic decision	Q_1 (%)	Δt_0 (day)	Most likely repair measure	Q_2 (%)	Δt_r (day)	Remarks
Complete (DS-5)	Concrete core crushing or rebar snapping at stem. Very large settlement of the approach slab.	δ_{step}: >28.7, or δ_{slope}: >64.47, whichever occurs first. Note: $\delta_{c,p}$: Settlement of step-offset approach slab (mm) $\delta_{c,r}$: Settlement of slope-offset approach slab (mm)	A. Fully open (normal operation) B. Mostly open (lane or speed restriction) C. Partially close (lane, speed and load restrictions) D. Fully close (complete closure)	A. 0 B. 0-20 C. 20-40 D. 40-60 E. 60-80 F. 80-100 G. 100	A. 0 B. 0-7 C. 7-14 D. 14-21 E. 21-30 F. 30-60 G. 60-90 H. 90-180	A. No need to repair B. Concrete surface sealing, filling, etc. C. Repaired by jacketing with steel plate, fiber reinforced plastic straps, etc. D. Repaired by additional reinforcement and concreting E. Demolish and rebuild F. Other (please detail in the last column as remarks)	A. 0 B. 0-20 C. 20-40 D. 40-60 E. 60-80 F. 80-100	A. 0 B. 0-7 C. 7-14 D. 14-30 E. 30-60 F. 60-90 G. 90-180 H. 180-360	

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Shear Key

Table 9. Survey of **Shear Key** restoration model variables.

Damage state	Qualitative damage description	Quantitative damage indices	(Please choose one answer from the provided options for each question)						
			Traffic decision	Q_1 (%)	Δt_0 (day)	Most likely repair measure	Q_2 (%)	Δt_r (day)	Remarks
(almost) Intact (DS-1)	Very minor crack at the intersection of the shear key and bent cap.	δ_{sk}: (0, 4] Note: δ_{sk}: Displacement at shear key top (mm)	A. Fully open (normal operation) B. Mostly open (lane or speed restriction) C. Partially close (lane, speed and load restrictions) D. Fully close (complete closure)	A. 0 B. 0-20 C. 20-40 D. 40-60 E. 60-80 F. 80-100 G. 100	A. 0 B. 0-7 C. 7-14 D. 14-21 E. 21-30 F. 30-60 G. 60-90 H. 90-180	A. No need to repair B. Concrete surface sealing, filling, etc. C. Repaired by additional reinforcement and concreting D. Demolish and rebuild E. Other (please detail in the last column as remarks)	A. 0 B. 0-20 C. 20-40 D. 40-60 E. 60-80 F. 80-100	A. 0 B. 0-3 C. 3-7 D. 7-14 E. 14-30 F. 30-60	
Slight (DS-2)	Slight crack at the intersection of the shear key and bent cap, but not throughout the shear key. Slight displacement at the top of shear key.	δ_{sk}: (4, 17.6] Note: δ_{sk}: Displacement at shear key top (mm)	A. Fully open (normal operation) B. Mostly open (lane or speed restriction) C. Partially close (lane, speed and load restrictions) D. Fully close (complete closure)	A. 0 B. 0-20 C. 20-40 D. 40-60 E. 60-80 F. 80-100 G. 100	A. 0 B. 0-7 C. 7-14 D. 14-21 E. 21-30 F. 30-60 G. 60-90 H. 90-180	A. No need to repair B. Concrete surface sealing, filling, etc. C. Repaired by additional reinforcement and concreting D. Demolish and rebuild E. Other (please detail in the last column as remarks)	A. 0 B. 0-20 C. 20-40 D. 40-60 E. 60-80 F. 80-100	A. 0 B. 0-3 C. 3-7 D. 7-14 E. 14-30 F. 30-60	

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Damage state	Qualitative damage description	Quantitative damage indices	(Please choose one answer from the provided options for each question)						
			Traffic decision	Q_1 (%)	Δt_0 (day)	Most likely repair measure	Q_2 (%)	Δt_r (day)	Remarks
Moderate (DS-3)	Crack at the intersection of the shear key and bent cap runs throughout the shear key. Concrete spalling is observed. mdoerate displacement at the top of shear key.	δ_{sk}: (17.6, 63.2] Note: δ_{sk}: Displacement at shear key top (mm)	A. Fully open (normal operation) B. Mostly open (lane or speed restriction) C. Partially close (lane, speed and load restrictions) D. Fully close (complete closure)	A. 0 B. 0-20 C. 20-40 D. 40-60 E. 60-80 F. 80-100 G. 100	A. 0 B. 0-7 C. 7-14 D. 14-21 E. 21-30 F. 30-60 G. 60-90 H. 90-180	A. No need to repair B. Concrete surface sealing, filling, etc. C. Repaired by additional reinforcement and concreting D. Demolish and rebuild E. Other (please detail in the last column as remarks)	A. 0 B. 0-20 C. 20-40 D. 40-60 E. 60-80 F. 80-100	A. 0 B. 0-3 C. 3-7 D. 7-14 E. 14-30 F. 30-60	
Severe (DS-4)	Multiple wide cracks run throughout the shear key, severe concrete spalling is observed, and rebars are visible. Severe displacemen at the top of shear key.	δ_{sk}: (63.2, 105.3] Note: δ_{sk}: Displacement at shear key top (mm)	A. Fully open (normal operation) B. Mostly open (lane or speed restriction) C. Partially close (lane, speed and load restrictions) D. Fully close (complete closure)	A. 0 B. 0-20 C. 20-40 D. 40-60 E. 60-80 F. 80-100 G. 100	A. 0 B. 0-7 C. 7-14 D. 14-21 E. 21-30 F. 30-60 G. 60-90 H. 90-180	A. No need to repair B. Concrete surface sealing, filling, etc. C. Repaired by additional reinforcement and concreting D. Demolish and rebuild E. Other (please detail in the last column as remarks)	A. 0 B. 0-20 C. 20-40 D. 40-60 E. 60-80 F. 80-100	A. 0 B. 0-3 C. 3-7 D. 7-14 E. 14-30 F. 30-60	

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Damage state	Qualitative damage description	Quantitative damage indices	(Please choose one answer from the provided options for each question)						
			Traffic decision	Q_1 (%)	Δt_0 (day)	Most likely repair measure	Q_2 (%)	Δt_r (day)	Remarks
Complete (DS-5)	Snapping of rebars in shear key, and extremely large displacement at the top of shear key, even collapse of the shear key.	δ_{sk}: >105.3 Note: δ_{sk}: Displacement at shear key top (mm)	A. Fully open (normal operation) B. Mostly open (lane or speed restriction) C. Partially close (lane, speed and load restrictions) D. Fully close (complete closure)	A. 0 B. 0-20 C. 20-40 D. 40-60 E. 60-80 F. 80-100 G. 100	A. 0 B. 0-7 C. 7-14 D. 14-21 E. 21-30 F. 30-60 G. 60-90 H. 90-180	A. No need to repair B. Concrete surface sealing, filling, etc. C. Repaired by additional reinforcement and concreting D. Demolish and rebuild E. Other (please detail in the last column as remarks)	A. 0 B. 0-20 C. 20-40 D. 40-60 E. 60-80 F. 80-100	A. 0 B. 0-3 C. 3-7 D. 7-14 E. 14-30 F. 30-60	

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Expansion Joint

Table 10. Survey of **Expansion Joint** restoration model variables.

Damage state	Qualitative damage description	Quantitative damage indices	(Please choose one answer from the provided options for each question)						
			Traffic decision	Q_1 (%)	Δt_0 (day)	Most likely repair measure	Q_2 (%)	Δt_r (day)	Remarks
(almost) Intact (DS-1)	No visible damage to expansion joint	Displacement within admissible displacement	A. Fully open (normal operation) B. Mostly open (lane or speed restriction) C. Partially close (lane, speed and load restrictions) D. Fully close (complete closure)	A. 0 B. 0-20 C. 20-40 D. 40-60 E. 60-80 F. 80-100 G. 100	A. 0 B. 0-7 C. 7-14 D. 14-21 E. 21-30 F. 30-60 G. 60-90 H. 90-180	A. No need to repair B. Replace accessories C. Replace the entire expansion joint D. Other (please detail in the last column as remarks)	A. 0 B. 0-20 C. 20-40 D. 40-60 E. 60-80 F. 80-100	A. 0 B. 0-7 C. 7-14 D. 14-21	
Complete (DS-2)	Apparently large displacement of the expansion joint	Displacement exceeding admissible displacement	A. Fully open (normal operation) B. Mostly open (lane or speed restriction) C. Partially close (lane, speed and load restrictions) D. Fully close (complete closure)	A. 0 B. 0-20 C. 20-40 D. 40-60 E. 60-80 F. 80-100 G. 100	A. 0 B. 0-7 C. 7-14 D. 14-21 E. 21-30 F. 30-60 G. 60-90 H. 90-180	A. No need to repair B. Replace accessories C. Replace the entire expansion joint D. Other (please detail in the last column as remarks)	A. 0 B. 0-20 C. 20-40 D. 40-60 E. 60-80 F. 80-100	A. 0 B. 0-7 C. 7-14 D. 14-21	

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Thank you very much for your patience to complete the survey. We sincerely appreciate your professional efforts on boosting the development of generalized component restoration models for highway bridges in China.

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